

ND-120 FPGA Debug Reference Card

Basys3 (xc7a35tcpg236-1) • 100 MHz • 115200 Baud • Branch: redo-idb

LED Layout (left LD15 to right LD0)

TERM LD15	CC3 LD14	CC2 LD13	CC1 LD12	CC0 LD11	— LD10	— LD9	— LD8	LCS LD7	MCLK LD6	BEAT LD5	UART LD4	RST LD3	RUN LD2	GRN LD1	RED LD0
LED	Signal		ON			OFF									
LD15	~TERM_n		Cycle termination active			Inactive									
LD14-11	~CC3..CC0_n		Cycle counter bits (should cycle rapidly)			-									
LD7	~LCS_n		Microcode loaded (~6ms after reset)			Still loading									
LD6	MCLK		Memory clock active			No memory clock									
LD5	clockTicks[26]		Blinks ~1.5Hz = clock OK			FPGA not clocked!									
LD4	~uartTx		UART transmitting			Silent									
LD3	sys_rst_n		Reset released (SW0 UP)			In reset									
LD2	~s_run		CPU executing			OPCOM mode									
LD1	cpu_led[1]		CPU GREEN			-									
LD0	cpu_led[0]		CPU RED (error)			Normal									

Switches

Switch	Signal	UP (1)	DOWN (0)
SW0 (left)	sys_rst_n	Reset released, CPU runs	CPU held in reset
SW1	btn2	7-seg = MAC address	7-seg = MIC address

ND-120 CPU • Norsk Data 1988 • FPGA Implementation • 2026-03-30

7-Segment Display (4 hex digits)

SW1	Shows	Range
DOWN	MIC address (CSA_12_0, microcode)	0000-1FFF
UP	MAC address (LA_23_10, memory)	0000-3FFF

Changing values = CPU executing • Stuck 0000 = CPU not running

UART

115200 Baud	8N1 Format	None Flow	A17 TX pin	A18 RX pin
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Boot Sequence

- 1 Program bitstream
- 2 LD5 blinks (clock running)
- 3 Slide SW0 UP (release reset)
- 4 LD3 ON (reset released)
- 5 LD15-11 cycling (state machine)
- 6 LD7 ON after ~6ms (microcode loaded)
- 7 LD2 ON (CPU running)
- 8 7-seg shows changing hex
- 9 Terminal shows # prompt

Vivado ILA Debug Signals (mark_debug)

Signal	W	Description
s_debug_csa[12:0]	13	MIC address (microcode fetch)
s_debug_la_23_10[13:0]	14	MAC address upper (LA 23:10)
s_debug_ca_9_0[9:0]	10	MAC address lower (CA 9:0)
s_debug_cc_term[4:0]	5	{TERM,CC3,CC2,CC1,CC0}_n
s_debug_mclk	1	Memory clock
s_debug_lcs_n	1	LCS_n (0=loading, 1=loaded)
s_run	1	CPU run (0=running)
s_debug_uartTx	1	UART TX line
s_debug_uartRx	1	UART RX line
s_debug_cpu_led[6:0]	7	CPU status LEDs

Deep Hierarchy (add via Set Up Debug)

Path	W	Description
CPU_BOARD/.../s_FIDB0_15_0	16	CGA data bus output
CPU_BOARD/.../s_FIDBI_15_0	16	CGA data bus input
CPU_BOARD/CPU/CS/s_csbits	64	TOPCSB microcode word
CPU_BOARD/.../s_idb_erb_out	16	Register file to IDB
CPU_BOARD/.../CHIP_32H/txState	3	UART TX state machine
CPU_BOARD/.../CHIP_32H/rxState	3	UART RX state machine
CPU_BOARD/CYC/s_cc0_n	1	Cycle counter bit 0